```
clear; clc; close all;

% --- Filter Specifications ---
Fs = 10000;          % Sampling Frequency (Hz)
N = 40;              % Filter Order
fp = 1500;           % Passband edge frequency (Hz)
fs = 2000;           % Stopband edge frequency (Hz)

% Normalized frequencies for design functions
f_vector = [0 fp fs Fs/2] / (Fs/2);
m_vector = [1  1  0    0];

% --- FIR Filter Design ---
% 1. Parks-McClellan (Equiripple) Design
b_pm = firpm(N, f_vector, m_vector);

% 2. Frequency-Sampling Design
b_fs = fir2(N, f_vector, m_vector);

% --- Frequency Response Calculation ---
[H_pm, f] = freqz(b_pm, 1, 2048, Fs);
[H_fs, ~] = freqz(b_fs, 1, 2048, Fs);

% ================================================
% DISPLAY NUMERICAL OUTPUTS IN COMMAND WINDOW
% ================================================
fprintf('=== FIR FILTER DESIGN COMPARISON ===\n\n');
fprintf('Filter Order (N)       : %d\n', N);
fprintf('Sampling Frequency (Fs): %d Hz\n', Fs);
fprintf('Passband Edge (fp)     : %d Hz\n', fp);
fprintf('Stopband Edge (fs)     : %d Hz\n\n', fs);

fprintf('--- PARKS-MCCLELLAN (EQUIRIPPLE) COEFFICIENTS (First 5) ---\n');
disp(b_pm(1:5));

fprintf('--- FREQUENCY-SAMPLING COEFFICIENTS (First 5) ---\n');
disp(b_fs(1:5));

% --- Plotting Results ---
figure('Name', 'Figure 1: Parks-McClellan vs Frequency-Sampling',
'NumberTitle', 'off');

% Subplot 1: Magnitude Response Comparison
subplot(2,1,1);
plot(f, 20*log10(abs(H_pm)), 'b-', 'LineWidth', 1.5); hold on;
plot(f, 20*log10(abs(H_fs)), 'r--', 'LineWidth', 1.5);
grid on;
xlim([0 5000]);
ylim([-80 10]);
title('FIR Filter Magnitude Response Comparison (Order N = 40)');
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
```

```
legend('Parks-McClellan (Equiripple)', 'Frequency-Sampling', 'Location',
'northeast');

% Subplot 2: Phase Response Comparison (Wrapped)
subplot(2,1,2);
plot(f, angle(H_pm)*180/pi, 'b-', 'LineWidth', 1.5); hold on;
plot(f, angle(H_fs)*180/pi, 'r--', 'LineWidth', 1.5);
grid on;
xlim([0 5000]);
ylim([-200 200]);
title('Phase Response Comparison');
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
legend('Parks-McClellan', 'Frequency-Sampling', 'Location', 'northeast');
```

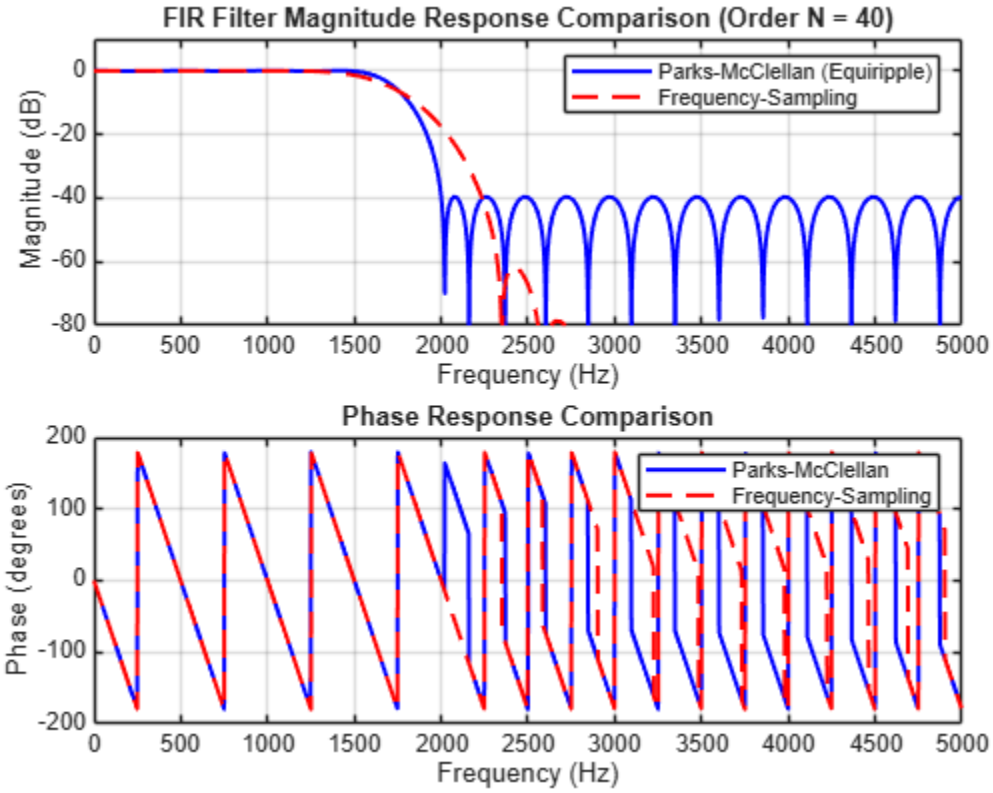
```
=== FIR FILTER DESIGN COMPARISON ===

Filter Order (N)       : 40
Sampling Frequency (Fs): 10000 Hz
Passband Edge (fp)     : 1500 Hz
Stopband Edge (fs)     : 2000 Hz

--- PARKS-MCCLELLAN (EQUIRIPPLE) COEFFICIENTS (First 5) ---
   -0.0015    0.0072    0.0053   -0.0010   -0.0081

--- FREQUENCY-SAMPLING COEFFICIENTS (First 5) ---
   1.0e-03 *

    0.0004    0.0751    0.1567   -0.0969   -0.7714
```



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